

OriginX: Continuous Conditioning of a Frozen Robot Policy with an Audited RoboCasa365 Evaluation

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Abstract

Adapting a strong robot policy with a small conditioning module is attractive, but a high aggregate evaluation score does not identify the benefit of that module. We present OriginX, an independent derivative of Xiaomi-Robotics-1-RoboCasa365 that combines an inherited Action LoRA adapter with an observation-conditioned continuous branch. The branch pools frozen visual-language features and proprioception, then adds a learned residual to the action transformer’s six adaptive-normalization conditioning channels. Only its 4,216,839 parameters are updated during the final training stage; stage labels supervise an auxiliary head and are excluded from inference. A frozen native-reset evaluation completes all 2,500 planned episodes across 50 RoboCasa365 tasks. OriginX succeeds in 1,496 episodes (59.84%), with 83.00%, 59.875%, and 33.75% on Atomic-Seen, Composite-Seen, and Composite-Unseen tasks. We retain every failure, the fixed seeds and horizons, and per-episode integrity records. However, an earlier 600-pair comparison against the original Xiaomi policy yields only a 1.00 percentage-point improvement, with a reported 95% interval of $[-2.17, 4.17]$ points and McNemar $p=0.6173$. The present evidence therefore establishes a complete author-run evaluation, not a reliable causal improvement or an organizer-validated rank. A subsequent, postselected study attempts paired replays of all 1,004 original failures with one observation-only Astra language intervention. It verifies 56 rescues, while retaining 813 non-rescues, 28 unknowns, and 107 initial-state deviations; the fixed-cohort verified rescue fraction is 5.58%. These conditional follow-up results leave the original 59.84% benchmark record unchanged. We describe both mechanisms, their evidence boundaries, and the controls still required for attribution.

1 Introduction

Generalist robot policies must combine perception, language understanding, and continuous control over long sequences. A model that reliably performs individual interactions can still fail when those interactions must be composed. Large pretrained vision-language-action policies provide a useful starting point, while their size makes unrestricted adaptation expensive and complicates attribution: gains may arise from additional data, an existing adapter, altered inference settings, or the proposed mechanism itself.

OriginX investigates a restricted adaptation interface. Rather than updating the full policy during its final stage, a small branch derives continuous conditioning from the policy’s existing observations. The branch has access to visual-language hidden states and recent proprioception. Its output modifies the conditioning already used by the frozen action transformer. An auxiliary stage objective supplies training supervision without requiring stage annotations at deployment. This construction provides a concrete hypothesis: observation-dependent conditioning could help a pretrained policy respond to intermediate task context while a teacher penalty discourages deviations from its inherited behavior. Neither the architectural construction nor that hypothesis implies a performance benefit.

We study this hypothesis using two distinct bodies of evidence. A completed 50-task evaluation measures the frozen endpoint under a fully specified native-reset protocol. A separate, earlier 30-task paired confirmation evaluates whether the candidate improves over the original Xiaomi policy in that development setting. The first is a single-arm measurement; the second does not support a reliable improvement. Combining them into a claim of superiority would be invalid because their task sets and evaluation protocols differ and because the new full evaluation lacks a matched control.

The contributions of this report are correspondingly limited. First, we specify the implemented continuous conditioning interface, including observation pooling, zero initialization, teacher anchoring, and the separation of training-only labels from inference. Second, we report all 2,500 planned outcomes together with evidence that preserves the evaluation denominator and

identifies the executed artifacts. Third, we retain the unsuccessful paired confirmation and report a separate, fixed-failure Astra follow-up with explicit intervention, eligibility, and usage accounting. These contributions provide a reproducible case study and a basis for further experiments; they do not establish a new state of the art. The work builds on Xiaomi’s released model.

2 Related Work

2.1 Generalist policies and household simulation

RoboCasa introduced diverse simulated household environments and tasks for studying generalist robot learning [1]. RoboCasa365 extends this setting and provides multi-task, foundation-model, and lifelong-learning benchmarks [2]. We use the multi-task evaluation’s Atomic-Seen, Composite-Seen, and Composite-Unseen splits in pretraining kitchen scenes [3]. The benchmark distinguishes training-task familiarity from compositional task generalization; our results should not be interpreted as covering all 365 tasks or all benchmark settings.

Xiaomi-Robotics-1 scales vision-language-action learning using large real-world trajectory collections followed by post-training for robot embodiments and instructions [4]. Its public RoboCasa365 checkpoint supplies the principal capabilities inherited by OriginX. Our small adaptation stage is not comparable in data or compute scope to training that foundation model. SIMPLER studies simulated evaluation of real-world manipulation policies and the importance of matching visual and control interfaces [8]. Its robot embodiments and protocols differ from the current experiment; no SIMPLER result is reported here.

2.2 Adaptation and continuous action generation

LoRA freezes pretrained weights while learning low-rank updates [5]. Our A2000 stage uses this established mechanism in the action model, and it must be treated as a separate inherited intervention rather than attributed to the later branch. Flow matching learns vector fields along specified probability paths [6]. OriginX retains the released policy’s flow-based action generator and adds conditioning to it; we do not introduce a new flow objective or integration algorithm. Diffusion Transformers demonstrate transformer-based generative modeling [7], providing relevant architectural context for the action transformer’s adaptive-normalization interface.

The teacher term is related to knowledge distillation’s use of teacher outputs as a learning target [9], but its role here is narrower: it penalizes disagreement with a frozen adapted policy at the same noisy action and flow time. It does not compress a stronger teacher into a

smaller standalone policy. The auxiliary stage head and the continuous modulation share a learned latent, yet no experiment presently isolates the value of stage supervision. Our method combines established ideas in a particular robot-policy interface; a broad claim of architectural novelty requires stronger comparison with prior adaptation and conditioning methods.

3 Method

3.1 Model lineage and notation

Let θ_0 denote the public Xiaomi-Robotics-1-RoboCasa365 checkpoint. A2000 adds rank-16, alpha-16 Action LoRA parameters α , producing an adapted policy $\theta_A = (\theta_0, \alpha)$. B2000 freezes the complete adapted policy, including its enabled LoRA parameters, and learns only branch parameters ϕ . The final OriginX endpoint is (θ_A, ϕ) . A2000 is a release alias for the unchanged adapter historically recorded as A1613; its actual training history remains 1,613 optimizer updates over 80,000 sampled windows. B2000 denotes the subsequent branch endpoint after 2,000 updates. Neither name denotes an independently trained foundation model.

For each batch item, let $H \in \mathbb{R}^{S \times D}$ be the hidden sequence from the original visual-language pass, with $D = 2560$. Let $s \in \mathbb{R}^{4 \times 60}$ be the padded state history. Only the first 14 state coordinates are valid in the released RoboCasa layout; coordinates 14 through 59 are verified as zero. The action model uses width $d = 1024$. Input validation preserves the released token, camera, temporal-group, and padding contracts.

3.2 Observation-only feature pooling

There are exactly three camera streams and four observed frames per stream. The processor groups the frames into two temporal patch groups. For camera c , the branch averages hidden states over spatial tokens in the most recent temporal group, producing $\bar{h}_c \in \mathbb{R}^D$. It also selects the last valid token after the video tokens, denoted h_{last} , retaining the released prompt ordering. The branch input is

$$x = [\bar{h}_1; \bar{h}_2; \bar{h}_3; h_{\text{last}}; \text{vec}(s_{:,0:14})] \in \mathbb{R}^{4D+56}. \quad (1)$$

The feature tensor is detached from the frozen policy. A 256-dimensional latent, stage logits, and a continuous conditioning residual are computed as

$$z = \text{SiLU}(W_e x + b_e), \quad \ell = W_s z + b_s, \quad (2)$$

$$\Delta_\phi = \text{reshape}(W_\Delta z + b_\Delta, 6, d). \quad (3)$$

The stage head has seven output classes. The output projection W_Δ and bias b_Δ are initialized to zero, so the

added modulation initially vanishes. The branch contains 4,216,839 trainable parameters, including its auxiliary head. The original frozen model has no separate failure detector, discrete option selector, external planner, or recovery controller. The later Astra study wraps this unchanged endpoint and is evaluated separately.

3.3 Conditioning the frozen action generator

The released action model projects its flow time into six conditioning channels. A bounded-scope forward hook adds Δ_ϕ to this time-projector output before its reshape into the six adaptive-normalization channels. The resulting conditioning is consumed by the native transformer layers. Symbolically, if $m_{\theta_A}(t)$ is the original time conditioning, then

$$\tilde{m}(t, o, s) = m_{\theta_A}(t) + \Delta_\phi(o, s). \quad (4)$$

We write the resulting velocity prediction as $v_{\theta_A, \phi}(a_t, t \mid o, s)$. This notation describes an additive conditioning intervention, not an additive correction directly to a motor command. The same observation-derived residual is available during the five Euler evaluations used for an action query. The implementation retains one visual-language pass per query and verifies the number of action-generation calls. Hooks are removed on both normal and exceptional exits, and reentrant requests are rejected.

Inference does not accept a true stage, phase label, stage confidence, or externally supplied stage residual. The seven logits are not used to select a hard class at deployment. Their shared latent influences the continuous residual, which participates in the actual action computation. Thus, the auxiliary target can affect learned representation during training without becoming privileged inference input.

3.4 Training objective

Let a be a demonstrated action chunk, ϵ the sampled flow noise, and $t \in [0, 1]$ the sampled flow time. The implementation forms $a_t = (1 - t)\epsilon + ta$ and uses $a - \epsilon$ as the velocity target. For sample i , let M_i be the conjunction of the action-coordinate mask and valid time-step mask. Define an equal-sample masked mean square,

$$\text{MSE}_{M_i}(r) = \frac{\sum_j M_{ij} r_{ij}^2}{\sum_j M_{ij}}. \quad (5)$$

Valid samples have nonempty masks. The flow loss is

$$L_{\text{flow}} = \frac{1}{B} \sum_i \text{MSE}_{M_i}(v_{\theta_A, \phi, i} - (a_i - \epsilon_i)). \quad (6)$$

The teacher is the frozen A2000 policy with its LoRA enabled and the new modulation absent. Student and

teacher receive the same noisy action, time, state embeddings, and cached visual-language context. The teacher is evaluated without gradients; checks reject mutation of shared inputs or consumption of additional random state. The retention loss is

$$L_{\text{ret}} = \frac{1}{B} \sum_i \text{MSE}_{M_i}(v_{\theta_A, \phi, i} - \text{stopgrad}(v_{\theta_A, i})). \quad (7)$$

For an available stage target y_i and detached annotation confidence q_i , the auxiliary objective is

$$L_{\text{stage}} = \frac{1}{B} \sum_i \mathbf{1}[y_i \geq 0, q_i \geq 1] \text{CE}(\ell_i, y_i). \quad (8)$$

The denominator is all batch windows, not only admitted labels. Low-confidence or missing stage labels suppress this auxiliary term without suppressing action supervision. The recorded objective is

$$L = L_{\text{flow}} + 1.0L_{\text{ret}} + 0.05L_{\text{stage}}. \quad (9)$$

This penalty encourages teacher consistency at sampled training states. It provides no theoretical guarantee against rollout regressions, and it may constrain useful changes. The strength of that trade-off remains an ablation question.

3.5 One-shot observation-only language assistance

After freezing the original results, we tested an external language wrapper without changing any model weights. Each original failed case receives a natural-reset control replay and an assisted replay with the same seed, task, original horizon, frozen B2000 endpoint, and inference settings. At the first 16-step policy query at or after half the horizon H , the assisted arm requests one language plan:

$$t_* = 16 \lceil H/32 \rceil. \quad (10)$$

This is a fixed schedule, not a learned failure or stall detector. The request includes the original task instruction, current left/right/wrist RGB images (each 256 by 256), remaining action-step budget, and routing metadata. Hidden simulator state, success predicates, original outcomes, and future frames are not supplied to the assistant. The frozen policy itself retains its ordinary proprioceptive inputs. The broker invokes gpt-6-astra through Codex CLI with high reasoning effort, batches at most eight independent requests into a labeled image contact sheet, and requires one schema-validated response per request. Tool calls are prohibited by the prompt and checked in the recorded event trace. The CLI command and completed trace document the requested model; the trace does not contain an independently signed model-weight identity.

The accepted subgoal text is limited to 1,500 characters. It is appended to the unchanged original instruction with an immediate-next-actions prefix and a reminder to finish the original task. Subsequent policy queries retain this augmented instruction. Astra neither emits low-level robot actions nor receives another observation for iterative correction. Simulation waits for the response, bounded by 1,200 wall-clock seconds; waiting does not increase the original environment-step horizon. Missing or invalid assistance remains a technical outcome. No gradient update, checkpoint selection, or outcome-based automatic retry is performed.

4 Data and Training

A2000 processes 80,000 sampled windows over 1,613 updates. Its batch size changed during training, so its final batch size cannot be multiplied by the entire update count to reconstruct its sample total. B2000 completes 2,000 updates with global batch size 128, or 256,000 sampled windows. These are sampled windows, potentially repeated, rather than distinct trajectories. B’s recorded learning rate is 10^{-4} , gradient-clipping threshold is 1.0, and elapsed training time is 8,453.98 seconds. The branch uses FP32 master parameters; the original model path uses the recorded mixed-precision arithmetic. We report the exact recorded configuration rather than infer unverified optimizer settings.

B samples Human300 replay and a pilot collection in equal proportions. The pilot contains 200 existing trajectories across 16 Composite-Seen tasks, split into 168 training trajectories and 32 held-out trajectories for diagnostic use. It uses existing annotation fields. No new human annotation, external reasoning-model annotation, or MimicGen data was added in this stage. The pilot split is a trajectory-level development split, not a substitute for the benchmark’s unseen task split.

The local sample-plan audit covers 80,000 A entries and 256,000 B entries, all marked as human data. Their task names are within Human300 and do not intersect the 16 official Composite-Unseen evaluation tasks. This audit is limited to the bound local adaptation plans. It does not independently establish the full data provenance of the inherited Xiaomi pretraining checkpoint. Data lineage, model lineage, and compute must therefore be reported separately. A planned C continuation was not executed, and no result in this paper depends on that proposed continuation.

5 Evaluation Protocol and Integrity

5.1 Fixed evaluation population

The final evaluation uses 50 target tasks: 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen,

in pretraining scenes. Each task receives exactly 50 episodes. The task-specific registered horizons from RoboCasa 1.0.1 are retained. Environment and policy seeds are fixed before rollout in the range 2090900000 through 2090902499. We do not replace seeds, discard completed failures, substitute easier tasks, or select another checkpoint in response to evaluation outcomes. The benchmark documentation defines the task and scene setting [3]; the author’s runner and implementation disclosures still require independent verification.

The frozen inference settings are five Euler steps, re-planning every 16 environment steps, observation history length four, observation interval two, and crop ratio 0.95. The protocol checks the original Counter reset implementation before and after every episode. This new native-reset run is distinct from earlier development experiments with reset-related implementation changes. Their outcomes are not inserted into its denominator.

5.2 Complete accounting

For task k with $n_k = 50$ episodes and success indicator Y_{ki} , the reported score is

$$\hat{S} = \frac{1}{50} \sum_{k=1}^{50} \frac{1}{50} \sum_{i=1}^{50} Y_{ki} = \frac{1496}{2500}. \quad (11)$$

The equality between macro task mean and pooled success rate follows from equal episode allocation. An unweighted mean of the three split percentages would instead overweight the smaller splits. All 2,500 planned trials have terminal records: 1,496 successes and 1,004 policy failures. Infrastructure-unknown outcomes and unattempted episodes are both zero. Each failed episode reaches its frozen original horizon. The record is complete for this manifest; completeness does not certify simulator equivalence or establish an official rank.

5.3 Execution and artifact checks

Simulation uses CPU OSMesa, rendering decimation supported by policy-query pixel checks, context-lifetime management, and readback guards. These are disclosed departures from a default graphics stack, not hidden implementation details. Eighteen frozen model services run across three H100 GPUs and serve up to 108 simulation clients with isolated random states. Parallelism changes execution throughput, not the definition of success. All owned model and episode processes terminate at campaign completion.

The audit verifies the hashes of all 2,500 result files and all 2,500 guard receipts, plus task membership, seeds, original horizons, reset checks, and aggregate counts. This records artifact identity and internal consistency. A hash alone does not establish scientific correctness. The campaign records approximately 3 hours 35 minutes including service initialization, and approximately 3 hours 13 minutes for the rollout phase.

Prior training and foundation-model development are excluded from these evaluation times.

5.4 Fixed-failure replay population and eligibility

The follow-up population is exactly the 1,004 failures of the original 2,500-episode record. It is postselected on baseline failure and is not an independent full-benchmark evaluation. The first execution partition claimed 36 cases; an infrastructure continuation processed the other 968. Whole-case exclusion of all 36 prior claims was frozen before continuation, regardless of success, failure, or technical outcome. Each case contributes once to the primary aggregate, with no cross-run fallback. The continuation adds read-only, same-socket RNG keepalives while awaiting language responses and permits dispatch to continue after recorded initial-state mismatches. It preserves episode logic, native reset, seeds, horizons, and intervention rule; these operational differences are retained in the protocol record.

A strict rescue R requires completed control and assisted episodes, a failed control and successful assisted outcome, matching initial-state evidence, exact preintervention state/image/instruction/action query hashes, and a verified applied Astra response. Let M count complete matched pairs with an exact preintervention prefix. Let E additionally require successfully applied assistance with valid request, response, image, and CLI evidence. Thus $R \leq E \leq M \leq 1004$. Initial-state deviations and missing or invalid episodes remain in the fixed denominator. Raw success bits are not a substitute for strict eligibility.

A four-case development pilot (assignment IDs 48, 97, 249, and 345) yielded zero strict rescues out of four and is reported separately. Its outcomes are never substituted into the primary aggregate. These four IDs remain in their uniquely assigned primary partitions, so they are development-exposed cases, not untouched holdouts. No new training was conducted for this study.

6 Results

6.1 Full native-reset evaluation

Split	Tasks	Successes	Episodes	Success rate
Atomic-Seen	18	747	900	83.00%
Composite-Seen	16	479	800	59.875%
Composite-Unseen	16	270	800	33.75%
Overall	50	1,496	2,500	59.84%

Table 1: Complete native-reset evaluation. Each task receives 50 trials.

The observed Atomic-Seen to Composite-Unseen gap is 49.25 percentage points. Composite-Unseen con-

tributes 530 of the 1,004 failures (52.79%). The five lowest-scoring tasks, all in that split, contribute 237 failures (23.61%): GatherTableware, HeatKebabSandwich, and PanTransfer each score 0/50; CategorizeCondiments scores 5/50; and SeparateFreezerRack scores 8/50. These are retrospective descriptive statistics, not separately selected test populations. Appendix B retains all tasks, including those with stronger performance.

The split comparison does not isolate the source of failure: task composition, object interactions, navigation, perception, horizon length, and training familiarity are not independently controlled. The original evaluation archive contains 10,048 JSON entries and no images or videos; initial-observation hashes cannot reconstruct the pixels. Episode-level success bits also do not reveal the first causal error in a trajectory. We therefore do not label uninspected failures as planning or recovery failures. The later assisted replays provide new observations; they are new runs and are not recordings of the original episodes.

At the October 9, 2026 check, the public leaderboard page marked updated October 8 displayed overall values of 58.1% for Paimon-0 and 57.4% for Xiaomi-Robotics-1 [10]. OriginX’s 59.84% is numerically higher than those displayed values. Those scores come from unmatched evaluations with no shared paired outcomes available here. The numerical difference is not a significance test or a validated first-place result. In particular, the published Composite-Unseen value for Paimon-0, 36.7%, exceeds OriginX’s 33.75%. We retain the public numbers as context rather than present them as a controlled comparison table.

6.2 Paired confirmation and negative evidence

An earlier exploratory evaluation on 30 tasks with five seeds each reported 72% for the original Xiaomi model and 76% for B2000. A subsequent confirmation used 20 new seeds per task, yielding 600 episodes for each model. It found the following outcomes:

Earlier paired confirmation	Original Xiaomi	OriginX B2000
Successful episodes	413/600	419/600
Success rate	68.83%	69.83%
B-only successes / original-only successes	47 original-only	53 B-only
Identical paired outcomes	500 ties	500 ties

Table 2: Earlier 600-pair confirmation on a separate development protocol; not a matched baseline for the final native-reset evaluation.

The paired difference is $(53-47)/600 = 0.0100$, or 1.00 percentage point. The retained analysis reports a 95% interval of $[-2.17, 4.17]$ points and McNemar $p=0.6173$.

These confirmation statistics are carried forward from the retained experiment report; the interval was not independently recomputed for this manuscript, and the full 600-pair records are separate from the released 2,500-episode archive. The predeclared confirmation requirements were a gain of at least five percentage points and a positive interval lower bound; neither requirement was met. This is insufficient evidence of improvement, not proof of equivalence. The exploratory gain cannot replace the independent confirmation result.

These 600 pairs belong to an earlier development setting, not to the final native-reset 50-task manifest. They must not be pooled with the 2,500 episodes or treated as a matched full-benchmark baseline. Moreover, the original-versus-B comparison changes both A2000 adaptation and the B branch. Even a positive result in that comparison would not isolate the branch’s effect. No matched full-protocol original-Xiaomi, A-only, zero-branch, or no-stage-supervision ablation has been completed.

6.3 What the evidence supports

The final record supports an observed success rate for a particular frozen model and protocol, with all planned outcomes accounted for. It supports a description of the implemented branch and its participation in inference. It does not show that the auxiliary stage objective improves action generation, that the teacher penalty preserves general capability, or that continuous modulation outperforms a same-budget alternative. The earlier confirmation’s uncertainty remains material even though the new single-arm aggregate is numerically competitive. Organizer acceptance and eligibility are separate questions under the public submission rules [11].

6.4 Completed conditional Astra follow-up

All 1,004 cases have a terminal classification and attempted-study coverage is complete. Valid experimental coverage is incomplete because unknowns and initial-state deviations remain. The unique aggregate combines 36 first-partition cases and 968 continuation cases, rather than selecting the better result from multiple executions.

There are $M = 874$ matched complete pairs and $E = 869$ verified applied-assistance pairs. The fixed-cohort verified rescue fraction is $R/1004 = 56/1004 = 5.58\%$. The conditional rates are $R/M = 6.41\%$ and $R/E = 6.44\%$. The raw matched table contains 56 assisted-only successes and 818 failures in both arms, with no control-only or both-arm successes. Five of those 818 pairs lack successfully applied, verified assistance and remain technical unknowns; they are not part of the 813 strict non-rescues. The 28 unknowns and 107 deviations are neither silently dropped nor called pol-

Fixed-failure classification	Cases	Fraction of 1,004
Strictly rescued	56	5.58%
Not rescued	813	80.98%
Technical unknown	28	2.79%
Initial-state deviation	107	10.66%
Total	1,004	100%

Table 3: Terminal classification of the postselected fixed-failure study. Percentages are rounded; unknowns and initial-state deviations remain in the denominator.

icy failures. These are descriptive conditional results; no full-population treatment effect or new benchmark score follows from them.

The rescued counts by original split are 12/153 Atomic-Seen failures, 25/321 Composite-Seen failures, and 19/530 Composite-Unseen failures. As one trace-verified example outside the four pilot IDs, OpenDrawer at seed 2090900186 has matching initial state and 24 exact preintervention queries. At step 384 of a 750-step horizon, Astra supplies a subgoal to approach the visible left drawer handle, grasp it, and pull outward. The assisted replay succeeds at step 674; its control fails at step 750. This documents a matched outcome and the applied instruction, not an independently established physical explanation of the rescue.

The settled usage ledger records 135 distinct primary-study CLI invocations: 134 successful calls and one error. Calls are deduplicated by invocation identity, not multiplied by the number of requests in a batch, and include failed or unapplied responses. Across the 134 calls with counters, known input tokens total 2,592,713, output tokens 195,237, and reasoning tokens 75,511; the error call lacks counters, so complete primary totals remain unknown. These counters are reported separately because reasoning tokens may overlap output accounting. Median measured CLI invocation latency is 46.16 seconds (range 2.33–77.35); this excludes queue delay and full rollout time. The pilot used one additional CLI invocation with 19,071 input, 629 output, and 137 reasoning tokens. Dollar cost is unknown for the authenticated CLI quota, not assumed zero.

The original frozen score remains 1,496/2,500 (59.84%). The 56 conditional rescues do not replace original failures, and summing them into that original score would mix protocols. The follow-up does not evaluate intervention-induced regressions on the 1,496 original successes.

7 Limitations and Research Agenda

7.1 Controlled experiments needed

A causal study should first evaluate the original Xiaomi checkpoint, A2000, and A2000 plus B under the same native-reset manifest and inference budget. A frozen zero-output branch provides an implementation-parity check; a separately trained branch without stage supervision tests the auxiliary objective. A no-retention arm tests teacher anchoring. Same-parameter or same-compute adaptation baselines are necessary before attributing gains to the continuous interface. Checkpoint selection should use development data, followed by a sealed final test manifest and all scheduled outcomes. Training-seed replication and task-aware paired intervals are needed to assess stability across optimization and task variation. These experiments are proposed and have not been executed for this manuscript.

The current data do not support a complete failure taxonomy. To develop adaptive recovery beyond the fixed half-horizon schedule tested here, a development-only failure study should annotate first visible errors and subsequent opportunities to recover. It should distinguish observation ambiguity, missed contact, wrong subgoal ordering, dropped objects, navigation errors, and time exhaustion only where trajectory evidence supports those labels. Multiple annotators or disagreement checks should quantify labeling reliability. Final evaluation episodes should remain untouched while hypotheses and intervention thresholds are developed. Training on the official Composite-Unseen tasks would change the benchmark’s task-generalization setting; retrospective inspection must not silently convert those tasks into training data. A future method developed from these findings needs a newly sealed evaluation with its data-access scope disclosed.

7.2 Astra attribution and generalization limits

An earlier offline gpt-6-astra agent inspected text records for CloseBlenderLid, PrepareCoffee, and GatherTableware and proposed falsifiable subgoal hypotheses. That analysis was exploratory and did not establish visual failure causes. The subsequent study in Sections 3.5, 5.4, and 6.4 actually executed observation-conditioned language interventions; its 56 strict rescues supersede the earlier absence of rescue measurements. Neither the offline hypotheses nor Astra’s generated rationales are ground-truth trajectory annotations.

The follow-up is a postselected fixed-failure study with a fixed half-horizon trigger, not a general recovery benchmark or evidence for a learned failure detector. It lacks same-budget deterministic instructions, generic instruction restatement, and alternative language-model con-

trols. These controls are needed to distinguish a benefit of Astra-specific reasoning from extra language conditioning or generic replanning. Unknowns, reset deviations, development exposure of four cases, and continuation infrastructure changes further limit interpretation. Even exact preintervention hashes and a failed control do not establish broad generalization from this selected population.

A stronger study should fix an assistance rule on development data and then run it on every episode of a newly sealed manifest, including cases that would otherwise succeed. It should report regressions, fixed denominators, missing evidence, uncertainty across tasks, language-model budget, latency, and token accounting. No result here establishes that assistance improves overall RoboCasa365 performance, that an adaptive trigger is effective, or that the branch itself causes the observed rescues.

7.3 Scope and publication readiness

The current results cover one endpoint, one simulator benchmark setting, and one inherited base policy. They do not establish robustness on physical hardware, cross-embodiment transfer, or SIMPLER performance. The released branch assumes three cameras and the RoboCasa state geometry; moving it to another robot interface is an adaptation problem. A scientific contribution strong enough for a competitive conference requires evidence beyond presentation: a precise new question, justified mechanism, matched gains or an informative negative finding, and independent reproducibility. The existing evidence does not justify a prediction of CVPR acceptance or spotlight selection.

8 Reproducibility, Attribution, and Responsible Use

The release materials include the adapter and branch weights, implementation snapshots, training-lineage records, evaluation configuration, manifest, all episode-result evidence, and a verification script. The base Xiaomi checkpoint is separately attributed and required for inference. Original evaluation sources preserve historical paths and execution assumptions. A later portable inference helper is packaging work, not the entry point used for the reported evaluation. Its CPU checks verify weight identities and mechanical contracts; GPU action parity and a new full-model rollout through that helper were not tested. These distinctions prevent packaging validation from being mistaken for a repeat of the benchmark.

Inference used Python 3.12, PyTorch 2.8.0 with CUDA 12.8, Transformers 4.57.1, and flash-attn 2.8.3. Simulation used Python 3.11, MuJoCo 3.3.1, robosuite 1.5.2, and RoboCasa 1.0.1. Full artifact identities and

environment details are retained in the released records. Reproduction should verify the reset implementation and action/preprocessing arithmetic before interpreting differences in rollout success. The final evidence archive identifies what was run; it does not eliminate the need for a clean-environment rerun.

This study evaluates simulated household behavior and makes no claim of safe physical deployment. Real-world use requires independent testing of actuation limits, contact behavior, perception errors, and stop conditions. The model is a derivative of public research artifacts, with upstream attribution and license notices retained. Astra supplied experimental language interventions as explicitly accounted above; automated tools also assisted code inspection and manuscript preparation; responsibility for the final claims, artifact provenance, and submission rests with the human author. Generated suggestions are not treated as experimental observations. The complete unsuccessful confirmation is disclosed to reduce selective reporting.

9 Conclusion

OriginX implements an observation-conditioned continuous branch over a frozen, adapted robot policy and completes a 2,500-episode RoboCasa365 evaluation at 59.84% success. The associated record preserves the evaluation population and the identities of the executed artifacts. An earlier paired confirmation does not establish a reliable improvement over the original policy, and the new full evaluation lacks the controls required to isolate the branch’s contribution. A separate one-shot Astra study verifies 56 rescues among the fixed 1,004 original failures, retaining all unknowns and reset deviations. This conditional evidence motivates controlled assistance studies while leaving the original 59.84% score and the unresolved branch-attribution question unchanged.

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A Evidence identities

The A2000 adapter has SHA-256 5892389863751f2245f83a6d3b96bf5725a052aa5b023230b093d8d8ece70ac4. The B2000 branch has SHA-256 41988b92391953687b39a0bc5a35bce962fbfcd08fbf6e550108430850d9944f. The final evaluation configuration has SHA-256 a3b0dbed4fb26ddcc472645fb9cc13008694293c23c49c325cc7524b4a93cd04; the manifest has SHA-256 cecee25e5750605c2c83643bca0fed55c8c4bfc46d672a934f0c2d29e760257. The per-episode evidence archive has SHA-256 4455049b4eae28214430b702ff20c9f47783dbe5c6966bff1ebe46c7e689ab18.

B Per-task outcomes

The following table is generated from the frozen evaluation summary. Every row has 50 trials. AS denotes Atomic-Seen, CS Composite-Seen, and CU Composite-Unseen. Horizons are environment steps. Task labels are preserved from the recorded manifest.

Task	Split	Horizon	Successes / 50
CloseBlenderLid	AS	900	18
CloseFridge	AS	900	50
CloseToasterOvenDoor	AS	450	44
CoffeeSetupMug	AS	600	33
NavigateKitchen	AS	450	41
OpenCabinet	AS	1050	47
OpenDrawer	AS	750	48
OpenStandMixerHead	AS	450	48
PickPlaceCounterToCabinet	AS	750	42
PickPlaceCounterToStove	AS	600	42
PickPlaceDrawerToCounter	AS	750	47
PickPlaceSinkToCounter	AS	900	47
PickPlaceToasterToCounter	AS	600	45
SlideDishwasherRack	AS	450	39
TurnOffStove	AS	750	34
TurnOnElectricKettle	AS	450	48
TurnOnMicrowave	AS	450	32
TurnOnSinkFaucet	AS	600	42
DeliverStraw	CS	2550	27
GetToastedBread	CS	3000	18
KettleBoiling	CS	1500	33
LoadDishwasher	CS	1800	37
PackIdenticalLunches	CS	3900	18
PreSoakPan	CS	2400	38
PrepareCoffee	CS	1800	11
RinseSinkBasin	CS	1350	32
ScrubCuttingBoard	CS	1200	37
SearingMeat	CS	4350	14
SetUpCuttingStation	CS	2400	38
StackBowlsCabinet	CS	2100	43
SteamInMicrowave	CS	2100	32
StirVegetables	CS	2400	27
StoreLeftoversInBowl	CS	2550	44
WashLettuce	CS	1650	30
ArrangeBreadBasket	CU	4350	35
ArrangeTea	CU	2250	23
BreadSelection	CU	1950	20
CategorizeCondiments	CU	1650	5
CuttingToolSelection	CU	1200	24
GarnishPancake	CU	2700	38
GatherTableware	CU	2250	0
HeatKebabSandwich	CU	2700	0
MakeIceLemonade	CU	3000	26
PanTransfer	CU	1800	0
PortionHotDogs	CU	2250	15

Task	Split	Horizon	Successes / 50
RecycleBottlesByType	CU	2850	12
SeparateFreezerRack	CU	2400	8
WaffleReheat	CU	4050	18
WashFruitColander	CU	3150	32
WeighIngredients	CU	3000	14

C Artifact availability

Identifying repository, release, and model-hosting links are deliberately excluded from this internal anonymous manuscript. They are maintained in the separate artifact note. No conference submission or acceptance is asserted.

The public release is a research artifact. Availability of files is distinct from organizer acceptance of the score and from publication acceptance. See the project status for live submission information; the numerical evaluation record above is fixed.